

Scenario Analyses: Effect of Utility Assumptions

METIMMOX-1 Economic Evaluation

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2025-11-10

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1 Overview

This scenario analysis examines the impact of utility assumptions on the cost-effectiveness of biomarker-guided immunotherapy strategies for metastatic MSS/pMMR colorectal cancer. We compare three scenarios:

1. **Base Case:** Standard utility values ($u_{np} = 0.9077$, $u_p = 0.9005$)
2. **Scenario 1 - Unadjusted Life Years:** All utilities set to 1.0 (no quality adjustment)
3. **Scenario 2 - Alternative Utilities:** Lower utility values ($u_{np} = 0.73$, $u_p = 0.59$)

For each scenario, we present deterministic cost-effectiveness results with ICERs and probabilistic sensitivity analysis results.

2 Base Case: Standard Utility Values

This analysis uses standard utility values derived from the IPD dataset ($u_{np} = 0.9077$, $u_p = 0.9005$).

Table 1: Base Case Utility Parameter Values

Parameter	Value	Description
u_{np} (Non-progressed utility)	0.9077	Utility value in non-progressed disease state
u_p (Progressed utility)	0.9005	Utility value in progressed disease state

Table 2: Base Case Cost-Effectiveness Results

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,748	1.804	NA	NA	NA	ND
crp	\$55,992	1.915	34244.12	0.1112157	\$307,907	ND
tmb_braf	\$68,032	1.823	NA	NA	NA	D
tlr	\$76,439	1.296	NA	NA	NA	D

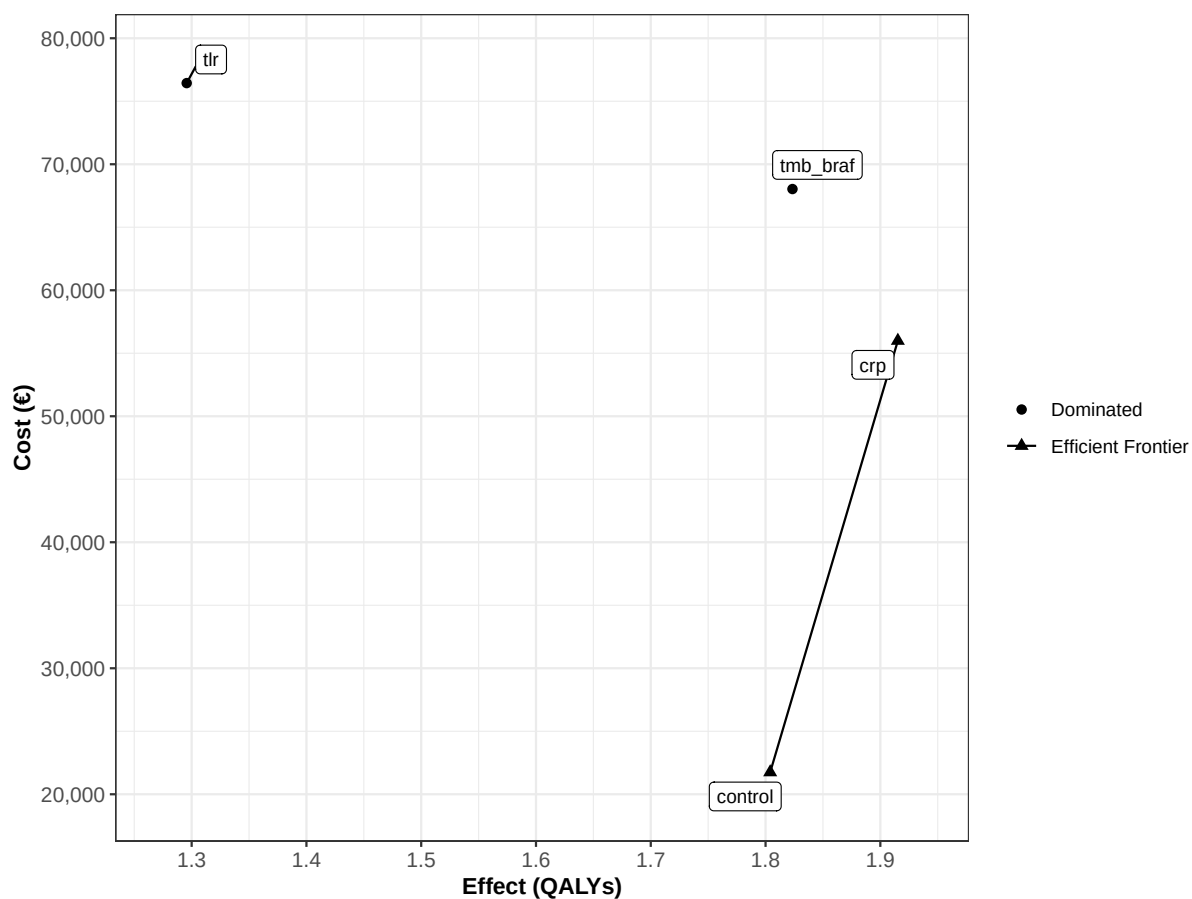


Figure 1: Cost-Effectiveness Plane - Base Case

Table: PSA Summary Statistics - Base Case

Table 3: PSA Summary - Base Case

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21799.20	1.80	2849.95	0.20
crp	55925.91	1.91	7373.45	0.21
tlr	76304.32	1.29	11615.49	0.14
tmb_braf	67933.36	1.82	9403.96	0.20

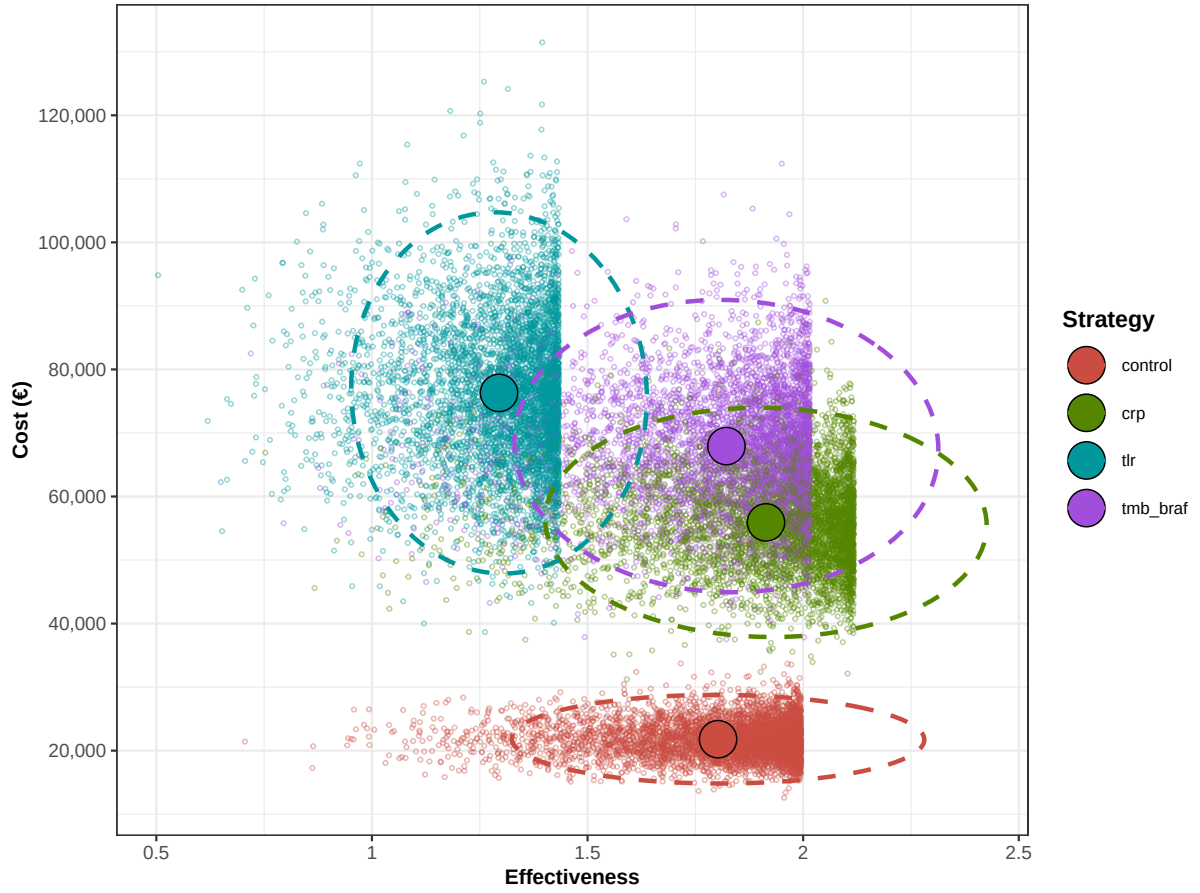


Figure 2: PSA Scatterplot - Base Case

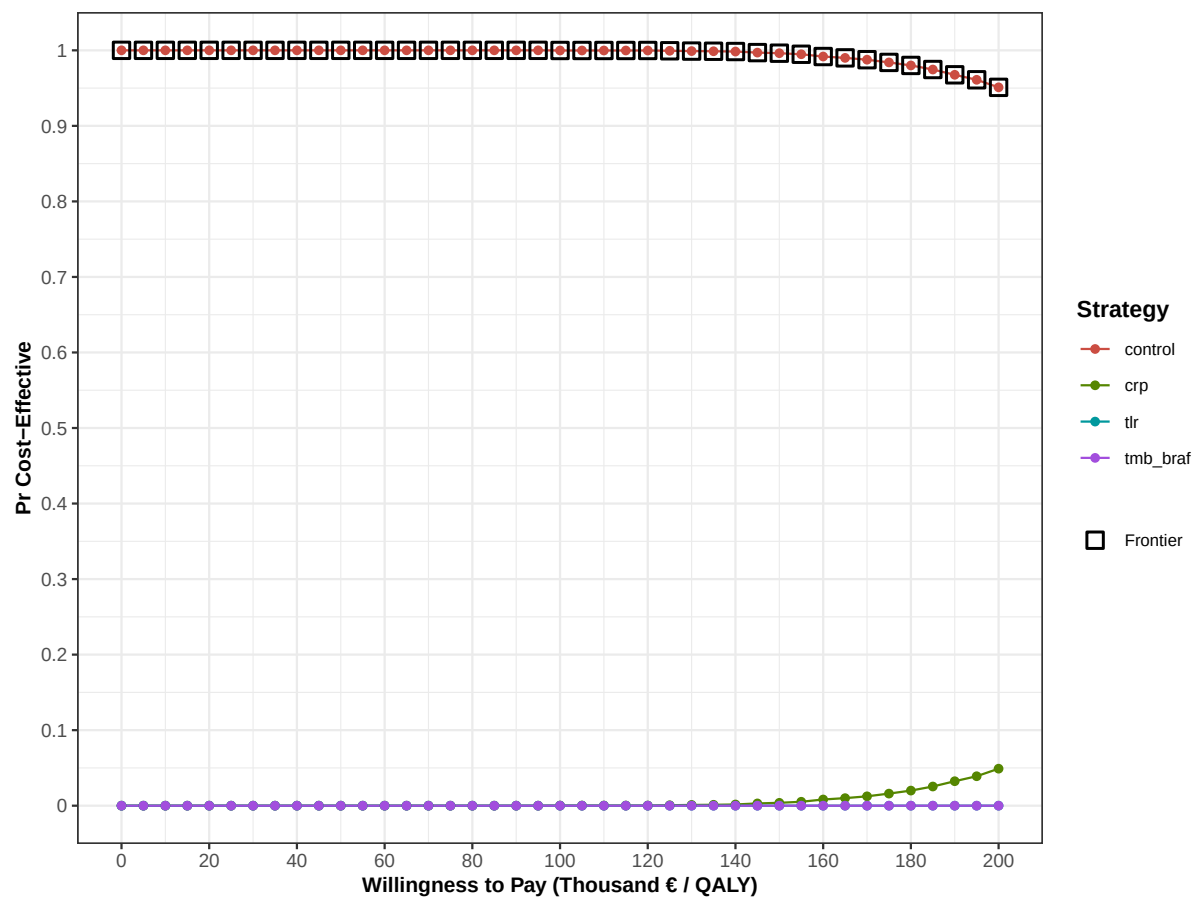


Figure 3: Cost-Effectiveness Acceptability Curve - Base Case

3 Scenario 1: Unadjusted Life Years (All Utilities = 1.0)

In this scenario, we analyze life years gained without quality-of-life adjustments, providing insight into the survival benefit independent of quality-of-life considerations.

Table 4: Utility Parameter Modifications for Scenario 1

Parameter	Base Case	Scenario 1	Difference
u_np (Non-progressed utility)	0.9077	1	0.0923
u_p (Progressed utility)	0.9005	1	0.0995

Table 5: Scenario 1 Cost-Effectiveness Results (Unadjusted Life Years)

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,748	1.995	NA	NA	NA	ND
crp	\$55,992	2.117	34244.12	0.1223059	\$279,987	ND
tmb_braf	\$68,032	2.015	NA	NA	NA	D
tlr	\$76,439	1.433	NA	NA	NA	D

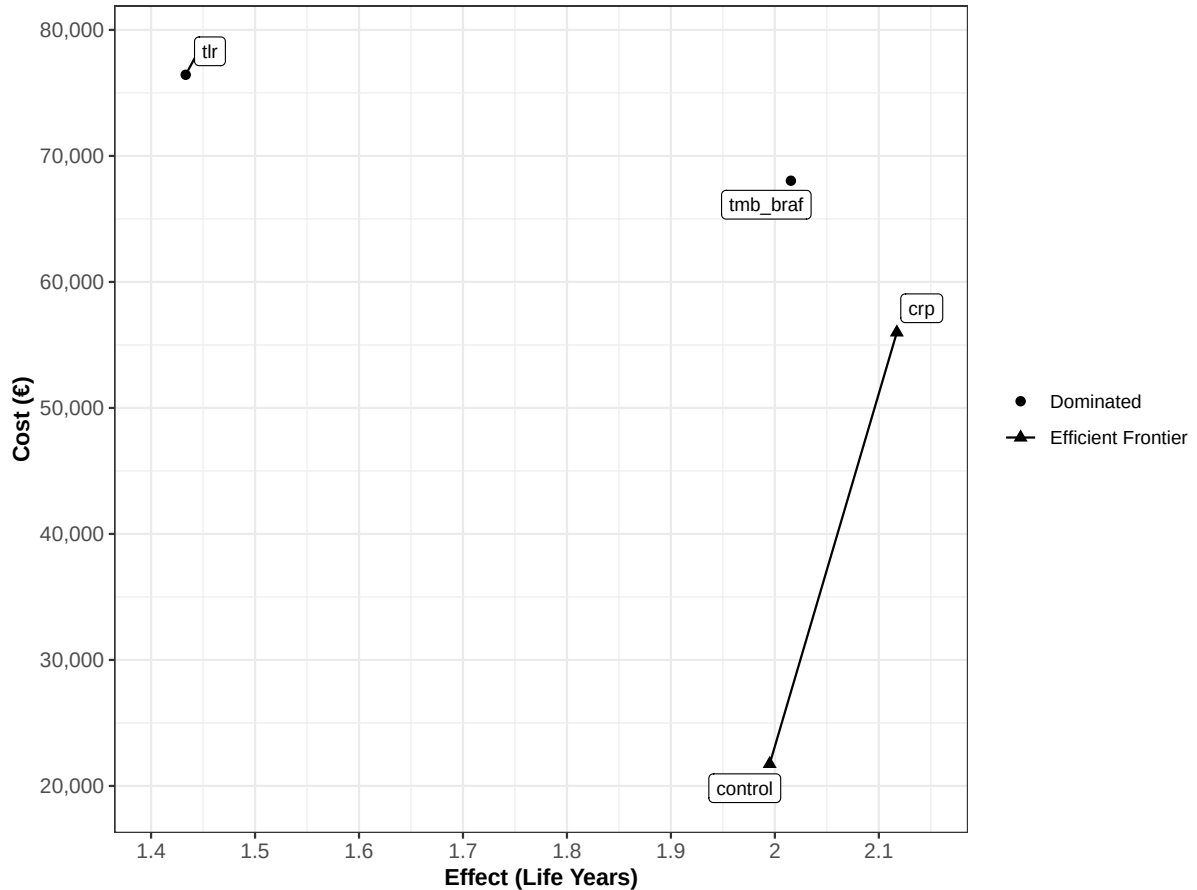


Figure 4: Cost-Effectiveness Plane - Scenario 1 (Unadjusted Life Years)

Table: PSA Summary Statistics - Scenario 1

Table 6: PSA Summary - Scenario 1 (Unadjusted Life Years)

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21708.70	2.00	2847.59	0
crp	55918.39	2.12	7591.36	0
tlr	76337.93	1.43	11896.82	0
tmb_braf	67943.44	2.02	9650.54	0

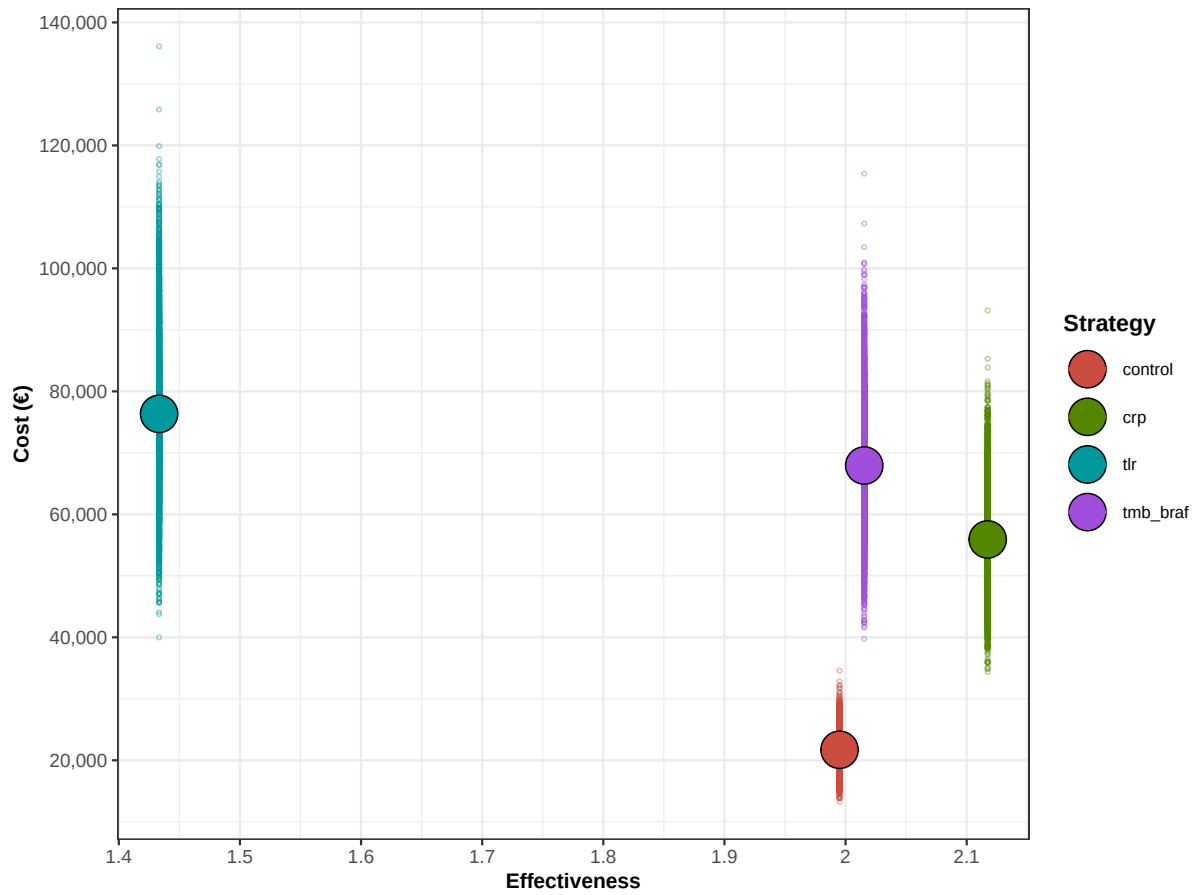


Figure 5: PSA Scatterplot - Scenario 1 (Unadjusted Life Years)

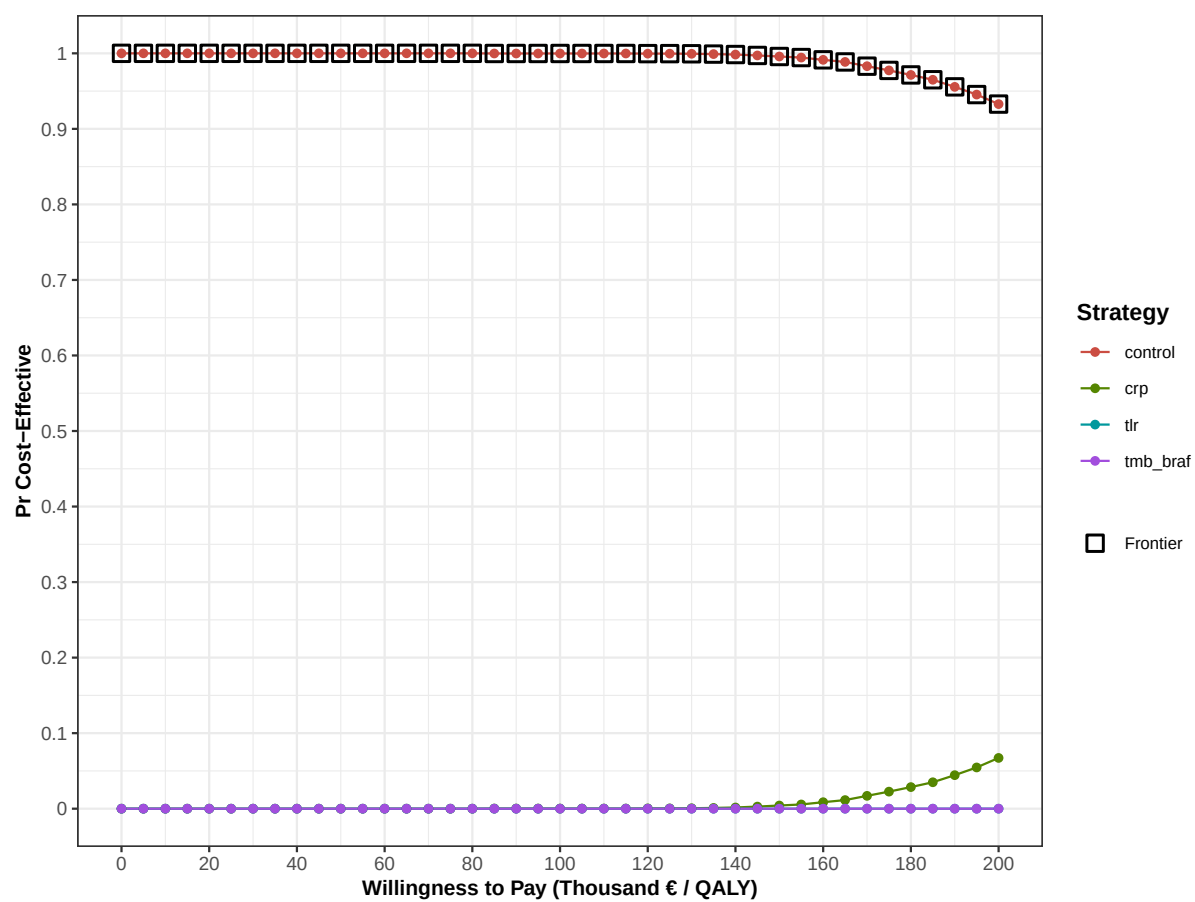


Figure 6: Cost-Effectiveness Acceptability Curve - Scenario 1

4 Scenario 2: Alternative Utility Values ($u_{np} = 0.73$, $u_p = 0.59$)

This scenario examines the cost-effectiveness under alternative, lower utility assumptions that may reflect more conservative estimates of quality of life. They are derived from the EQ-5D index score of the CORRECT trial (Grothey et al. Lancet 2013, as cited by Gourzoulidis et al. J Comp Effect Res 2018)-

Table 7: Utility Parameter Modifications for Scenario 2

Parameter	Base Case	Scenario 2	Difference	% Change
u_{np} (Non-progressed utility)	0.9077	0.73	-0.1777	-19.5770
u_p (Progressed utility)	0.9005	0.59	-0.3105	-34.4808

Beta Distribution Parameters for PSA:

Table 8: Beta Distribution Parameters for Scenario 2 PSA

Parameter	Mean	CV	Alpha	Beta
u_{np}	0.73	0.15	11.270	4.168
u_p	0.59	0.15	17.632	12.253

Table 9: Scenario 2 Cost-Effectiveness Results (Alternative Utilities)

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,748	1.323	NA	NA	NA	ND
crp	\$55,992	1.416	34244.12	0.0931452	\$367,642	ND
tmb_braf	\$68,032	1.356	NA	NA	NA	D
tlr	\$76,439	0.943	NA	NA	NA	D

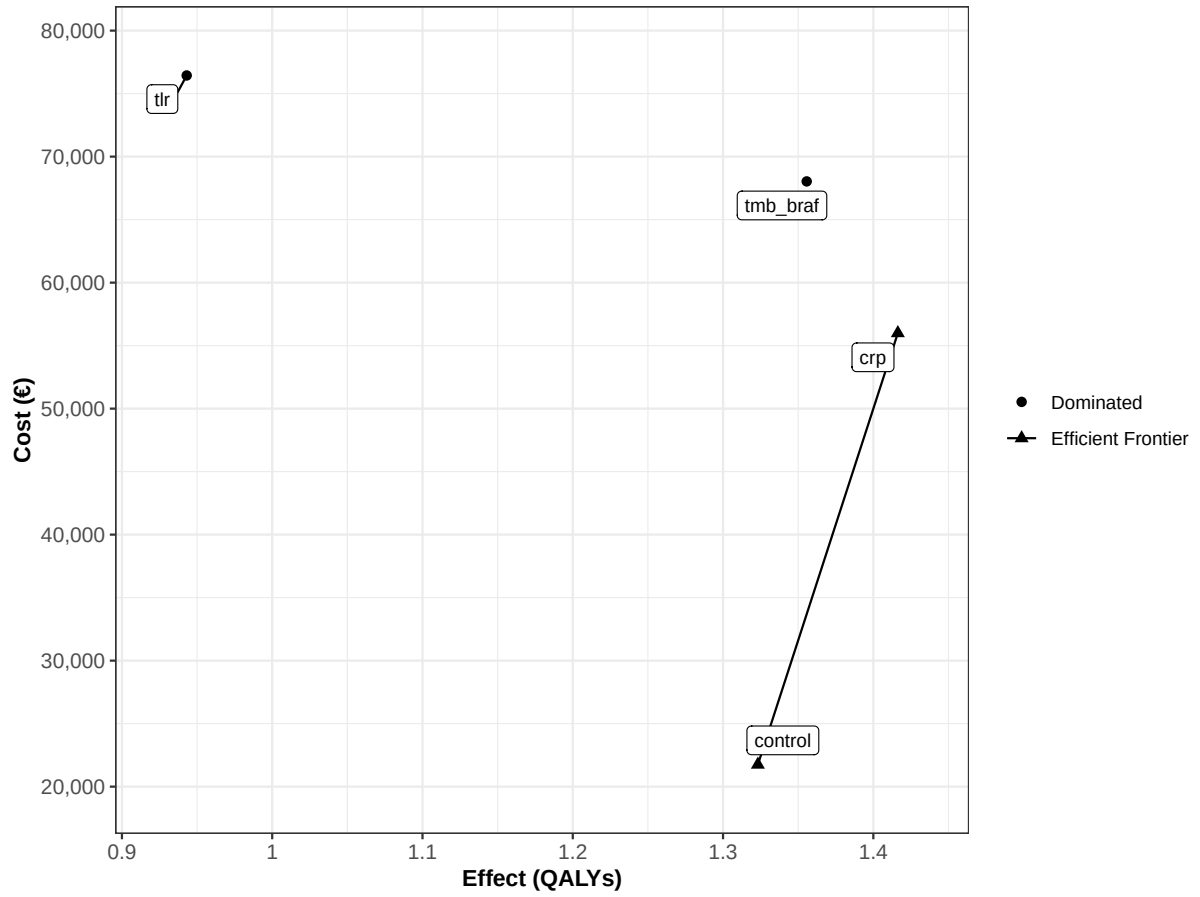


Figure 7: Cost-Effectiveness Plane - Scenario 2 (Alternative Utilities)

Table: PSA Summary Statistics - Scenario 2

Table 10: PSA Summary - Scenario 2 (Alternative Utilities)

Strategy	meanCost	meanEffect	sdCost	sdEffect
control	21733.94	1.32	2910.31	0.14
crp	56086.72	1.41	7506.58	0.15
tlr	76603.47	0.94	11794.94	0.10
tmb_braf	68161.13	1.35	9554.64	0.15

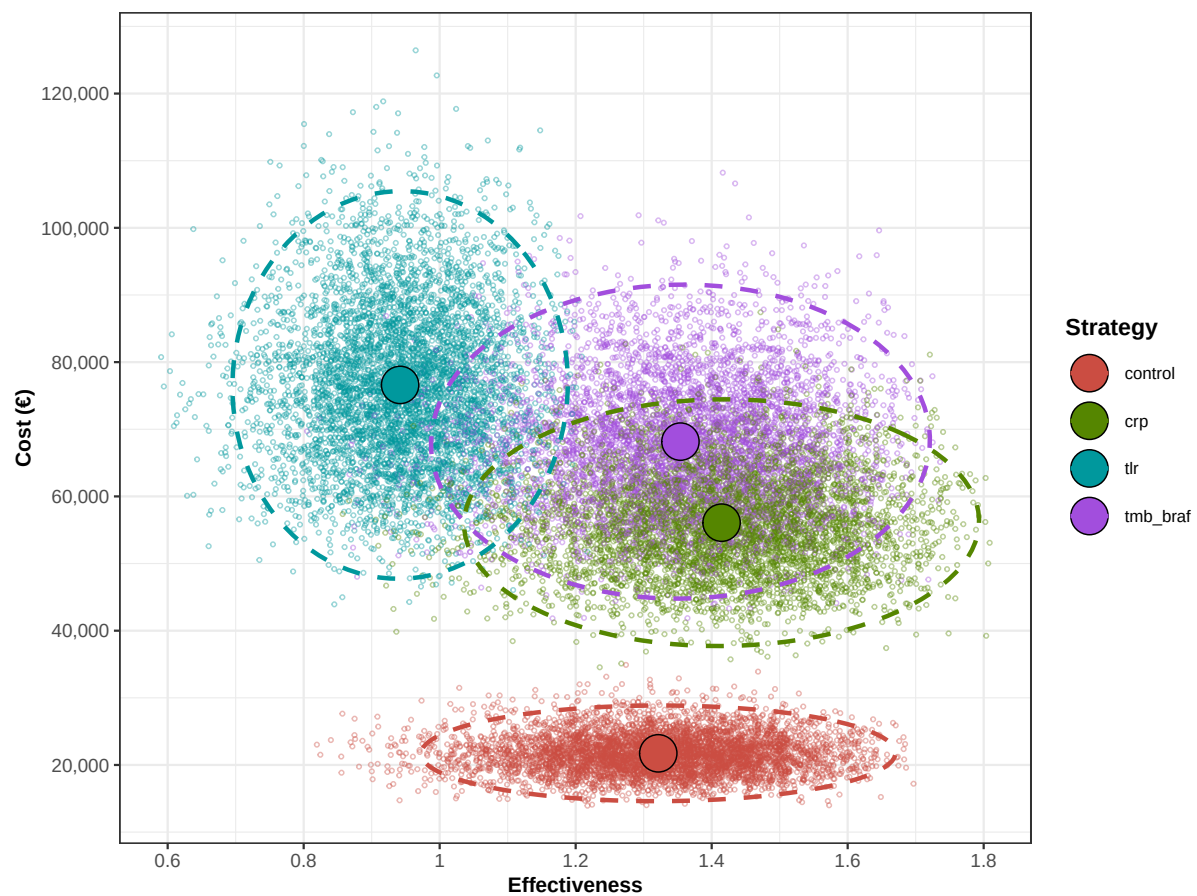


Figure 8: PSA Scatterplot - Scenario 2 (Alternative Utilities)

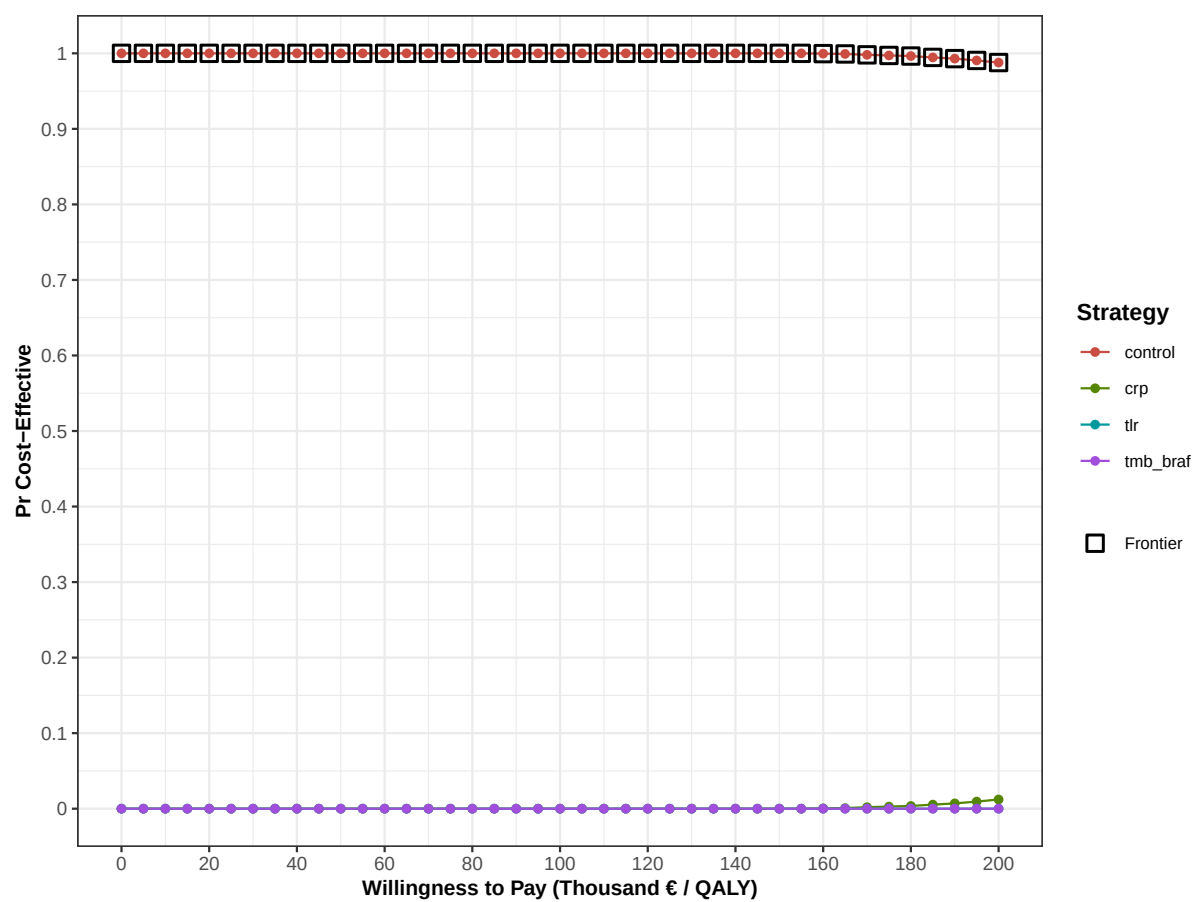


Figure 9: Cost-Effectiveness Acceptability Curve - Scenario 2

5 Comparative Analysis Across All Scenarios

Table 11: Utility Parameter Values Across Scenarios

Scenario	u_np	u_p	Description
Base Case	0.9077	0.9005	Standard utility values from literature
Scenario 1 (Life Years)	1.0000	1.0000	No quality adjustment (all utilities = 1.0)
Scenario 2 (Alternative)	0.7300	0.5900	Lower/conservative utility estimates

Table 12: Cost-Effectiveness Results Comparison Across Scenarios

Scenario	Strategy	Cost (€)	Effect	NMB (€)
Base Case	Standard of Care	\$21,748	1.804	\$70,258
	CRP-guided	\$55,992	1.915	\$41,686
	TLR-guided	\$76,439	1.296	-\$10,360
	TMB/BRAF-guided	\$68,032	1.823	\$24,964
Scenario 1	Standard of Care	\$21,748	1.995	\$79,998
	CRP-guided	\$55,992	2.117	\$51,992
	TLR-guided	\$76,439	1.433	-\$3,342
	TMB/BRAF-guided	\$68,032	2.015	\$34,754
Scenario 2	Standard of Care	\$21,748	1.323	\$45,733
	CRP-guided	\$55,992	1.416	\$16,239
	TLR-guided	\$76,439	0.943	-\$28,342
	TMB/BRAF-guided	\$68,032	1.356	\$1,107

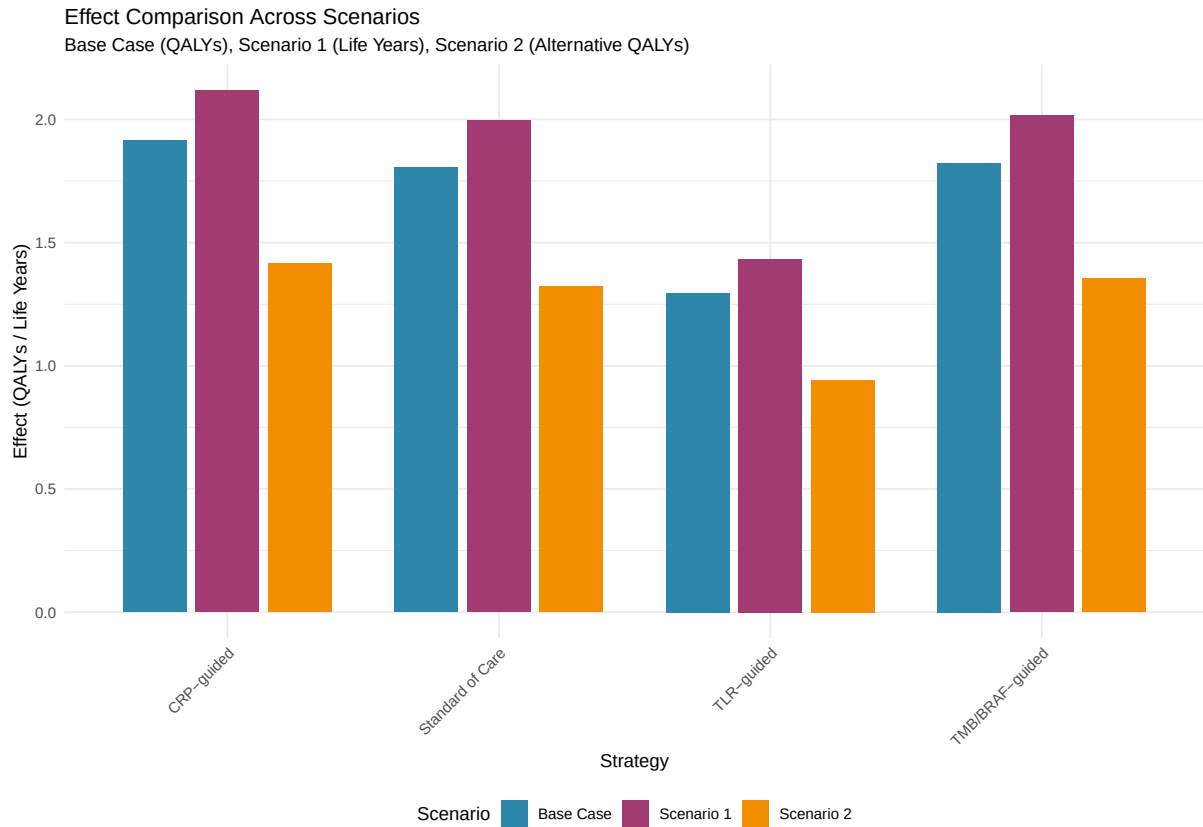


Figure 10: Comparison of Effects Across Scenarios by Strategy

Table 13: ICER Comparison Across Scenarios (€ per QALY/Life Year)

Strategy	Base Case	Scenario 1	Scenario 2
Standard of Care	Reference	Reference	Reference
CRP-guided	\$307,907	\$279,987	\$367,642
TMB/BRAF-guided	Reference	Reference	Reference
TLR-guided	Reference	Reference	Reference

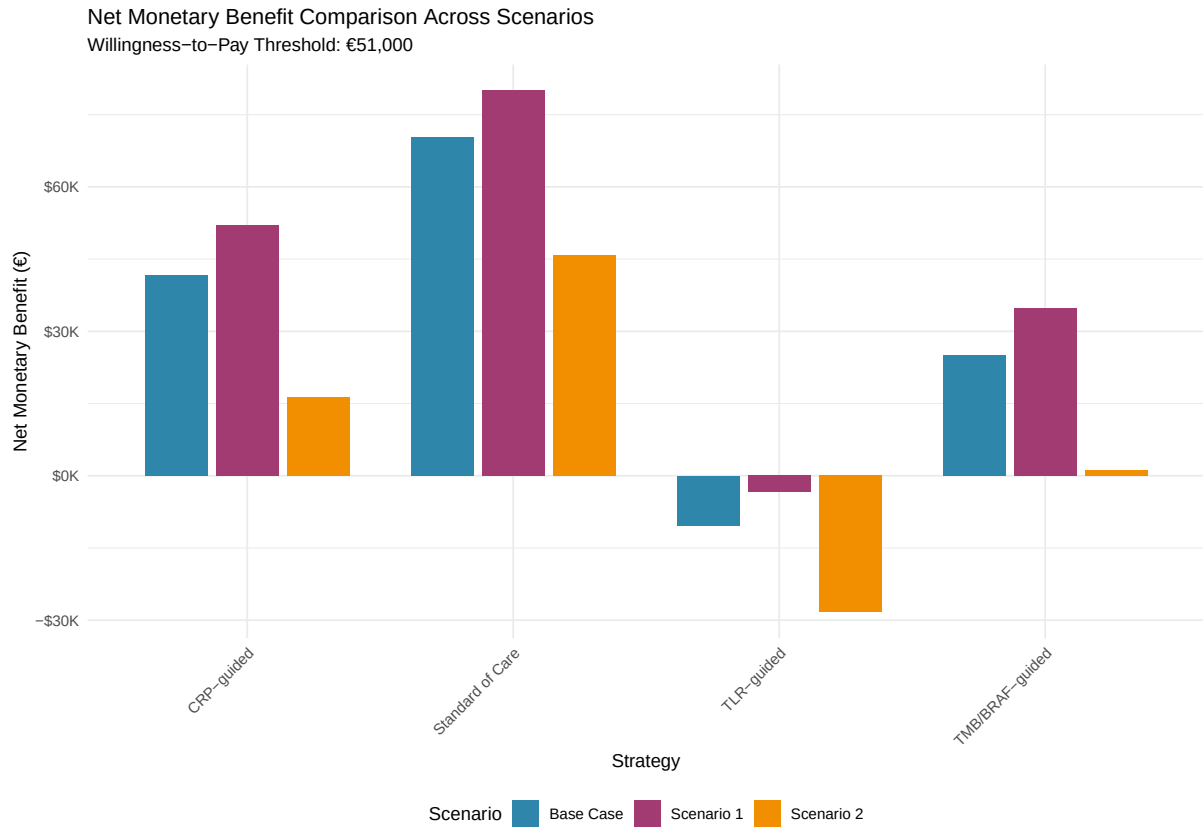


Figure 11: Net Monetary Benefit Comparison Across Scenarios (WTP = €51,000)

Table 14: Optimal Strategy by Scenario at WTP Threshold

Scenario	Optimal Strategy	WTP Threshold
Base Case	Standard of Care	\$51,000
Scenario 1 (Life Years)	Standard of Care	\$51,000
Scenario 2 (Alternative)	Standard of Care	\$51,000

6 Summary

This scenario analysis examined the impact of utility assumptions on the METIMMOX+1 model. The cut-off effectiveness from the base case scatter plot disappears when using alternative values.

Report completed on: 2025-11-10

Repository: [ben-geisler/METIMMOX-1](#)

Report version: 1.0